

Sayan Banerjee

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Summary

Data-driven professional with experience across the full data lifecycle, combining analytics with product thinking to drive business outcomes. Adept at transforming complex datasets into actionable insights, enabling informed decision-making and user-focused improvements. Strong collaborator with a focus on solving ambiguous problems and delivering meaningful, measurable outcomes.

Experience

Data Analyst, WorkIndia, Bangalore

May 2025 - Present

- Increased **Daily Active Users (DAU)** by **17%** through behavioral cohort segmentation and targeted outreach strategy.
- Designed controlled A/B experiments on 1M daily website traffic with equal test-control allocation over 14 days, achieving an **8% lift in Clicks-to-Leads at 99% statistical confidence**.
- Led continuous A/B tests across Email and WhatsApp channels, boosting downloads by 34% and 26% YoY.
- Improved **Cost-per-DAU from 14.5 to 9.7 (33% reduction)** and saved **\$1,200 in server costs** by designing time-affinity notification strategies using historical interaction data.
- Wrote complex SQL queries (joins, aggregations, window functions) to analyze retention trends, recency-frequency metrics, and campaign performance across segmented user cohorts.
- Built and maintained cron-driven ELT workflows using SQL and PySpark to automate segmentation pipelines and KPI tracking dashboards.
- Ensured reporting accuracy and data integrity in production environments, proactively mitigating risks from ETL and workflow failures.

Technical Research Analyst, Hevo Data, Bangalore

Jun 2024 - Dec 2024

- Investigated and resolved issues across **80+ distributed ETL pipelines**, improving data reliability for downstream analytics and business reporting.
- Created a Snowflake-based pricing tool used by marketing teams, contributing to a **9% increase in qualified leads**.
- Performed log-level and metric-driven root cause analysis to support system stability and safeguard reporting accuracy.

Research and Development Intern, Samsung Prism, Remote

Nov 2023 - May 2024

- Processed and analyzed **32,000+ documents** using scalable data pipelines for structured information extraction.
- Developed transformer-based summarization models, balancing model accuracy, throughput, and deployment efficiency.

Education

KIIT University, B.Tech in Computer Science

Sep 2021 - Apr 2025

- **CGPA:** 8.9/10.0
- **Coursework:** Machine Learning, Database Management Systems, Distributed Systems, Cloud Computing

Skills

- **Programming & Analytics:** Python, SQL, PySpark, Pandas, NumPy
- **Data Analysis:** A/B Testing, Cohort Analysis, KPI Tracking, Statistical Validation, EDA, Hypothesis Testing, Experiment Design
- **Databases:** MySQL, PostgreSQL, Snowflake, MongoDB, Athena
- **Tools & Infrastructure:** Git, Linux, JIRA, Confluence, Kubernetes, Jenkins

Academic Projects

Perplexa, KIIT 

Jan 2025 - Apr 2025

- Integrated Google OAuth with MongoDB-backed authentication and Captcha verification, ensuring **99.9% uptime**.
- Implemented a low latency FAISS-based RAG solution with real-time web context, **reducing hallucination by 7%**.
- Deployed via SSL-secured GitHub webhooks to Streamlit Cloud, scaling to **500 concurrent queries/min**.

Vehicle Movement Analysis and Insight Generation, Intel Unnati Industrial Training 

May 2024 - Jul 2024

- Leveraged YOLOv8 for real-time vehicle detection and tracking, achieving **<100ms inference latency** and **95% accuracy**.
- Used PaddleOCR for license plate recognition, processing 60+ frames/min with **92% accuracy** under variable conditions.
- Optimized edge inference using OpenVINO, realizing a **3x speed boost** for efficient traffic monitoring.

Text-to-SQL, KIIT 

Dec 2023 - Feb 2024

- Fine-tuned Star-Coder2-3b on SQL data using PEFT LoRA, achieving a **28% improvement** in query-to-code accuracy.
- Reduced input processing time by 35% by introducing vectorized computation and caching token embeddings.
- Applied 4-bit quantization using bitsandbytes, **lowering memory usage by over 20%**.